

Navier - Stokes equations

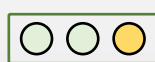
$$\frac{\partial u_i}{\partial t} + u_i \cdot \nabla u_i = -\frac{1}{\rho_i} \nabla p_i + \nu_i \nabla^2 u_i + \delta_i \quad \nabla \cdot \mathbf{u}_i = 0$$

NS Equation Variant 1



Lid-Driven Cavity Flow

$\vec{\theta}_1$



Network 1

NS Equation Variant 2



Pipe Flow

$\vec{\theta}_2$



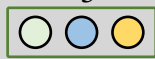
Network 2

NS Equation Variant 3



Couette Flow

$\vec{\theta}_3$



Network 3

Different **variants** of the same PDE with **few changes in Parameters and Boundary Conditions.**

Almost completely different neural network models

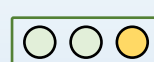
(a) Single-task PINNs

NS Equation Variant 1

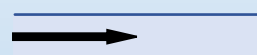


Lid-Driven Cavity Flow

$\vec{\theta}_1$



NS Equation Variant 2



Pipe Flow

$\vec{\theta}_2$

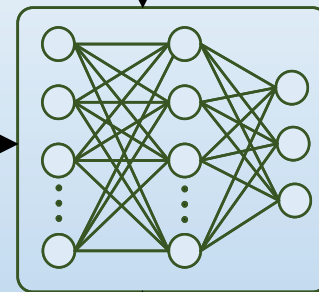
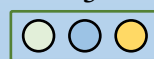


NS Equation Variant 3



Couette Flow

$\vec{\theta}_3$



Unified PINN framework for Multi-task learning of diverse Navier-Stokes equations

(b) Unified PINN Framework